

Rethinking Sleep Staging: Unsupervised cross-laboratory mouse sleep staging with temporal variational autoencoders and substage discovery

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ABSTRACT

Cross-laboratory manual scoring of mouse sleep from electroencephalography (EEG) and electromyography (EMG) lacks consensus, and supervised models trained at one site often fail elsewhere without retraining or per-animal calibration. To stage held-out animals without that calibration, we study twenty quality-screened wild-type mice from three European Mouse Sleep Staging Validation (MSSV) laboratories under leave-mice-out cross-validation with four folds. Using spectral EEG and EMG alone as input, decoder-only conditional variational autoencoders (cVAE) assign states through the encoder and a learned mixture prior at inference. When the prior adds bout structure via a hidden Markov model–Gaussian mixture model (HMM–GMM), mean prior normalized mutual information (NMI) reaches 0.575 on joint cross-laboratory holdout versus 0.282 for a matched static mixture, a gain of 0.29 NMI averaged over folds. At finer mixture counts, macro NMI and prior log-density diverge on holdout fold 4 and on the full cohort. Population prior log-density is highest at $K=8$ among completed runs, and we propose a map of three Wake substates, two NREM cores, consolidated REM, and one short transition bucket. Figs 4–6 summarise the map. Expert labels enter only for post-hoc evaluation. The method enables label-free cross-laboratory staging and comparison of proposed substage structure before expert labels are harmonised.

AUTHOR SUMMARY

Sleep biologists score mouse EEG and EMG into Wake, NREM, and rapid eye movement (REM) sleep, but raters at different laboratories often disagree. We trained unsupervised models on mice from Bern, Copenhagen, and Lyon, then tested them on held-out animals the models had never seen, without calibrating each new recording by hand. Our best model reads spectral brain and muscle signals through a neural network and enforces realistic sleep bout structure with a temporal prior. Our aim was to stage sleep without expert labels and to propose substages from the data. It outperformed simpler baselines on cross-laboratory holdout and recovered an eight-substage population map whose spectra and bout lengths matched patterns we discussed with sleep neuroscientists at the Kornum laboratory, indicating structure beyond Wake, NREM, and REM alone.

1 INTRODUCTION

Manual mouse sleep scores disagree across MSSV laboratories, and supervised models trained at one site fail on held-out animals elsewhere unless retrained or calibrated per mouse [1–5]. Expert staging nonetheless maps continuous physiology into Wake, NREM, and REM, the standard three-class benchmark. Unsupervised work in mice has resolved substages with distinct spectra, bout lengths, and transition roles that Wake/NREM/REM labels alone do not name [6]. Because montage and laboratory protocols differ across studies [7], claims that sleep structure extends beyond macro scoring must be tested on new animals and new sites under population holdout without per-animal calibration at inference.

Unsupervised mouse EEG/EMG workflows can discover latent states from hand-crafted or learned features, often with HMM-style temporal structure [8], but rarely under leave-mice-out holdout across laboratories in a shared population model. Katsageorgiou et al. resolved rodent substages beyond the macro taxonomy with distinct physiology and bout structure in mcRBM [6], yet without cross-laboratory leave-mice-out holdout. By contrast, García Ciudad et al. showed that longer temporal context improves supervised rodent staging, especially for REM [9], but that setting remains fully supervised and does not test unsupervised cross-site transfer. In our MSc thesis [10], we developed mixture-VAE staging on Copenhagen mice alone with encoder+decoder conditioning and expert-engineered features. That work reached similarly high prior NMI under the same $K=3$ macro grid of about 0.7 NMI. This report uses decoder-only conditioning and a spectral CNN front-end under cross-laboratory leave-mice-out holdout, a strictly harder evaluation. Expert labels are used for post-hoc evaluation only.

We extend decoder-only cVAE with mixture priors: a static GMM for cGMVAE [11], decoder conditioning [12], and a sticky HMM-GMM sequence prior composed with the encoder [13]. This type of model has not been applied on our sleep data before. Subject embeddings condition reconstruction during training only, not the encoder mapping from spectra to latents. The encoder therefore represents sleep-state structure shared across animals, while the decoder learns site-specific amplitude and montage signatures needed for reconstruction. Held-out mice are staged from encoder means and the learned prior without subject identifiers. We first validate zero-shot cross-laboratory holdout at $K=3$, then ask whether latents support finer structure than Wake, NREM, and REM through mixture-resolution sweeps and physiology-first interpretation [6, 10]. On locked spectral inputs from laboratories 2, 3, and 5 at Bern, Copenhagen, and Lyon, we compare a static Gaussian mixture model (GMM) prior with a warm conditional HMM-GMM variational autoencoder (cHMM-GMVAE) prior. Holdout fold 4 is the primary generalisation benchmark for macro metrics, whereas the full cohort supports substage physiology interpretation in Results (Figs 5 and 6) and supplementary PCA panels. An overview roadmap linking analyses, figures, and appendix panels is in Appendix E.1. Dataset and protocol details are in Section 2 and Appendix A (Tables A.1 and A.4).

2 DATA

Dataset and cohort. MSSV on OpenNeuro ds006366 [7] comprises invasive EEG and neck EMG from 92 mice across five European laboratories. Expert Wake, NREM, REM, and Artifact labels are provided on a 4 s grid for evaluation only. We analyse laboratories 2 at Bern, 3 at Copenhagen, and 5 at Lyon. These are healthy wild-type sites with parietal-frontal invasive EEG and neck EMG comparable across sites. We exclude laboratory 1 (Bologna), which uses a single ipsilateral fronto-parietal differential rather than separate parietal and frontal contacts, and laboratory 4 (Copenhagen pathology cohort), which uses prefrontal-cortical montages not matched to the Bern/Lyon layout [7]. Quality screening removes mice with unstable low-frequency EEG and weak EMG Wake/NREM contrast before modelling. Drift is quantified as relative 0.5–2 Hz power ≤ 0.2 of 0.5–30 Hz on each retained electrode. EMG must separate Wake from NREM with a Wake/NREM root-mean-square ratio ≥ 2 . The screened cohort comprises 20 mice with 6, 10, and 4 per site and ~ 453 h of recording (per-mouse inventory in Appendix A.1). We use four-fold leave-mice-out cross-validation at the mouse level (fold assignments in Appendix A.4). Lyon contributes only four quality-screened mice, which caps the fold count while keeping every fold a strict subject holdout. Training mice never appear in the validation fold for a given split. No new animal procedures were performed.

Preprocessing and inputs. Montages are harmonised to a parietal-frontal EEG differential plus neck EMG. Bern (laboratory 2) records four invasive EEG channels, two parietal and two frontal, following MSSV cohort B [7]. Copenhagen and Lyon each use one parietal and one frontal contact (EEG1 and EEG2). At Bern we use EEG1 (parietal) and EEG4 (frontal) so the pairing matches the parietal-frontal layout at laboratories 3 and 5. That pair was fixed from incohort ablation over the alternative frontal contact EEG3. Locked passbands follow incohort winners: Bern EEG 0–30 Hz and EMG 3–100 Hz without post-normalisation, Copenhagen and Lyon EEG 0–20 Hz and EMG 5–60 Hz with run-wise post-normalisation (Appendix C.3, Table C.3, Appendix A.3). Each recipe was fixed from incohort ablations on training mice in that laboratory before any holdout run, not tuned on validation folds. Recordings follow the MSSV 4 s annotation grid at 128 Hz sampling. For cross-validation holdout runs, expert-labeled Artifact epochs are removed before

training. Each epoch is converted to a log-power spectral map $\mathbf{x}_t \in \mathbb{R}^{C \times F}$ rather than raw voltages, because time-domain HMM baselines stay near chance on this cross-lab protocol (Appendix B). Batched epochs form contiguous sequences for temporal models.

3 METHODS

3.1 MODEL AND TRAINING

Spectral EEG/EMG epochs are mapped to a low-dimensional latent space by a cVAE. Subject embeddings condition reconstruction during training only, not encoder inference. Staging at holdout uses encoder means and a learned mixture prior without subject identifiers, so latents must encode cross-animal sleep structure rather than site-specific nuisance absorbed only in the decoder. We compare three deep baselines on matched inputs and cross-validation folds. cGMVAE uses a static GMM with $T=1$. cHMM-GMVAE uses a warm HMM-GMM sequence prior with $T=64$ epochs, about 4.3 min, and $K=3$ for staging. HMMGMVAE uses the same CNN-VAE and warm HMM-GMM prior as cHMM-GMVAE but disables subject embeddings, isolating decoder-only conditioning while holding the sequence prior fixed. All three share the same locked per-laboratory spectral preprocessing and training recipe (Appendix B.1), with three seeds per fold. Table 1 contrasts cGMVAE with cHMM-GMVAE for the temporal prior and HMMGMVAE with cHMM-GMVAE for decoder-only subject conditioning at matched $T=64$. cGMVAE uses $T=1$ because the static mixture scores epochs independently, whereas HMMGMVAE and cHMM-GMVAE use $T=64$ windows for the sequence prior.

Design choices. Inputs are log-power spectra with a CNN front-end because raw time-domain HMMs fail to generalise across mice on MSSV. Latent dimension $d=6$ matches the thesis-scale mixture models, sequence length $T=64$ gives about 4.3 min of bout context, and sticky self-transition weight $\kappa=0.92$ encourages persistent sleep states. Fold 4 sensitivity checks showed shorter windows without the warm schedule collapsed holdout NMI (Appendix B). cHMM-GMVAE warm-starts from a Gaussian encoder, then a static mixture, then ramps in the HMM-GMM prior to avoid cold-start collapse. Training minimises mean squared reconstruction on log-power maps with a ramped β KL weight. Checkpoints are selected without expert labels from validation log-likelihood and state-occupancy entropy (Appendix B), and prior NMI is computed only after that selection.

Encoder-decoder. Training batches are contiguous windows $\mathbf{x}_{1:T} \in \mathbb{R}^{T \times C \times F}$ with $T = 64$. A convolutional encoder maps each epoch to a diagonal Gaussian posterior,

$$q_{\phi}(\mathbf{z}_t | \mathbf{x}_t) = \mathcal{N}(\mathbf{z}_t | \boldsymbol{\mu}_t, \text{diag}(\boldsymbol{\sigma}_t^2)), \quad (\boldsymbol{\mu}_t, \log \boldsymbol{\sigma}_t^2) = f_{\phi}(\mathbf{x}_t), \quad (1)$$

with f_{ϕ} a CNN-multilayer perceptron (MLP) encoder and $\mathbf{z}_t \in \mathbb{R}^d$, $d=6$. Latent samples use the reparameterisation $\mathbf{z}_t = \boldsymbol{\mu}_t + \boldsymbol{\sigma}_t \odot \boldsymbol{\epsilon}_t$ with $\boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The decoder reconstructs each epoch from $\mathbf{z}_t$ and, during training only, a subject embedding $\mathbf{e}_i \in \mathbb{R}^8$:

$$\hat{\mathbf{x}}_t = g_{\theta}(\mathbf{z}_t, \mathbf{e}_i). \quad (2)$$

Under decoder-only conditioning, f_{ϕ} does not receive $\mathbf{e}_i$. CNN and MLP widths are in Appendix B.1. Fig 1 summarizes the training and holdout staging path.

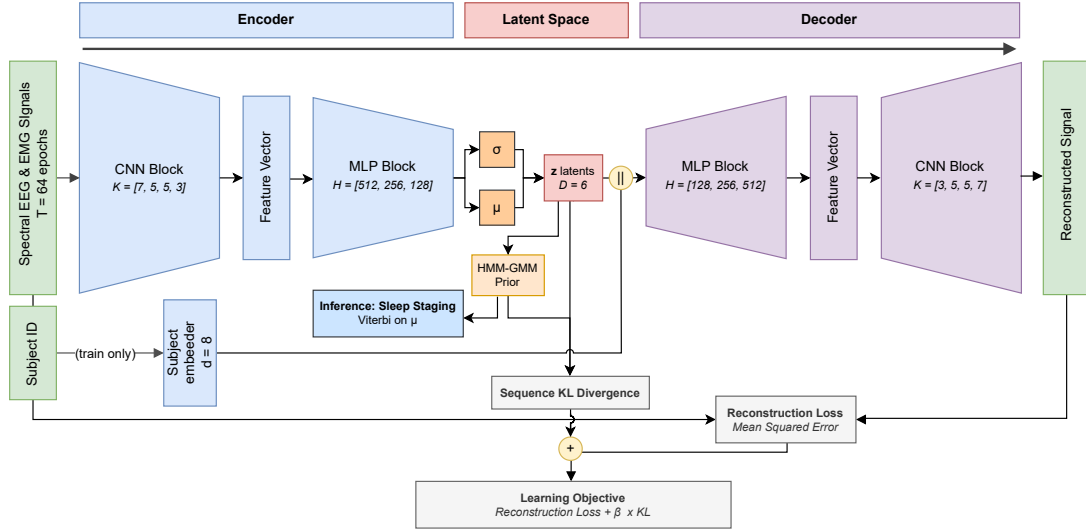


Fig 1. Conditional Gaussian mixture variational autoencoder (GMVAE) architecture. Encoder: spectral EEG/EMG maps pass through CNN and MLP blocks. Subject ID feeds a subject embedder used only in the decoder. Latent: diagonal Gaussian posterior with dimension $d=6$ feeds a GMM (cGMVAE) or HMM-GMM (cHMM-GMVAE) prior. Decoder: latent vector concatenated with subject embedding, then CNN reconstruction. Training minimises mean squared error (MSE) reconstruction plus $\beta \times$ prior KL. At holdout, only the encoder and prior are used for staging.

Training objective. For each window $\mathbf{x}_{1:T}$, training minimises

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \beta \mathcal{L}_{\text{KL}}, \quad (3)$$

where

$$\mathcal{L}_{\text{recon}} = \frac{1}{T} \sum_{t=1}^T \|\mathbf{x}_t - \hat{\mathbf{x}}_t\|_2^2 \quad (4)$$

is mean squared reconstruction error over T contiguous 4s epochs. The Kullback-Leibler (KL) term averages the log-density ratio between the encoder and the mixture prior over the window,

$$\mathcal{L}_{\text{KL}} = \frac{1}{T} \sum_{t=1}^T \log q_{\phi}(\mathbf{z}_t | \mathbf{x}_t) - \log p(\mathbf{z}_{1:T}), \quad (5)$$

where $\log p(\mathbf{z}_{1:T})$ is the joint prior marginal over the full sequence (not repeated per epoch). For cHMM-GMVAE, $\log p(\mathbf{z}_{1:T})$ is the HMM-GMM forward marginal with emission densities $b_k(\mathbf{z}_t)$ and transition matrix A . For cGMVAE, each epoch is scored independently with a static GMM marginal at $T=1$. The weight β is zero for the first ten epochs, then ramps linearly from 0.01 to 1.0. cHMM-GMVAE warm-starts the mixture prior from encoder means (Appendix B.4). Phase timings and prior equations are in Appendix C.1 and Appendix B.4.

Prior-based staging at holdout. Held-out mice are staged from encoder means μ_t and the trained prior without subject embeddings or decoder reconstruction. Using means rather than sampled latents reduces Viterbi variance across seeds once the encoder is trained. Static cGMVAE assigns $\hat{s}_t = \arg \max_k [\log \pi_k + \log b_k(\mu_t)]$. cHMM-GMVAE applies Viterbi decoding,

$$\hat{\mathbf{s}}_{1:T} = \arg \max_{\mathbf{s}_{1:T}} \left[\log \pi_{s_1} + \sum_{t=2}^T \log A_{s_{t-1}, s_t} + \sum_{t=1}^T \log b_{s_t}(\mu_t) \right], \quad (6)$$

with $B_{tk} = \log b_k(\mu_t)$.

3.2 EVALUATION PROTOCOL

The twenty quality-screened mice were partitioned into four leave-mice-out folds (Table A.4, Appendix A.4). Joint scope pools all non-held-out mice from laboratories 2, 3, and 5 into one population model and tests on the held-out fold across sites. Within-lab scope restricts both training and validation to a single laboratory with the same fold assignments. Fold 4 is the primary benchmark for macro metrics and the holdout K -sweep because it balances held-out load across sites after moving one Copenhagen mouse out of that fold (Appendix A.4). The population K -sweep uses the same locked recipe on all twenty mice but retains Artifact epochs in the training stream. Macro NMI still compares predicted states to Wake, NREM, and REM only. Per-laboratory spectral locks remain fixed at holdout. Agreement with expert Wake, NREM, and REM is prior NMI,

$$\text{NMI}(Y, \hat{Y}) = \frac{2I(Y; \hat{Y})}{H(Y) + H(\hat{Y})}, \quad (7)$$

where I is mutual information and H Shannon entropy. We use NMI as the primary metric because predicted mixture indices are arbitrary up to permutation, whereas macro-aligned accuracy first maps each cluster to a dominant expert stage. NMI scores partition agreement without that alignment step and remains defined when mixture count exceeds three, but its interpretation weakens when expert labels stay three-class. Mixture count is selected from prior log-density instead (Substage resolution and physiology). Each fold trains three seeds. Holdout prior NMI uses the best unsupervised checkpoint per seed and reports the maximum across seeds, matching incohort ablation reporting but optimising relative to seed means (Limitations). Macro-aligned accuracy, Cohen's κ , decoder-only ablation, and substage sweep setup are in Appendix B and C.

Data and code availability. MSSV is available on OpenNeuro as ds006366. Key hyperparameters are listed in Appendix B.1. Processed features and analysis code accompany this report.

4 RESULTS AND DISCUSSION

4.1 CROSS-LABORATORY HOLDOUT AT MACRO RESOLUTION

We evaluated decoder-only cGMVAE and cHMM-GMVAE under joint leave-mice-out training on twenty quality-screened mice from Bern, Copenhagen, and Lyon (Table 1). cGMVAE, HMMGMVAE, and cHMM-GMVAE were trained with the same locked preprocessing and CNN-VAE hyperparameters per laboratory, differing in prior type, sequence length ($T=1$ versus $T=64$), and subject-embedding use. cGMVAE versus cHMM-GMVAE tests the temporal prior, and HMMGMVAE versus cHMM-GMVAE tests decoder-only subject conditioning at matched $T=64$. Although supervised pipelines on MSSV reach much higher accuracy when retrained per site or given long temporal context [2, 3, 9], we report unsupervised performance under that matched recipe. cHMM-GMVAE reached mean prior NMI 0.575 versus 0.282 for cGMVAE, a gain of 0.29 averaged over four folds.

Table 1. Joint holdout prior NMI by fold (best-of-three seeds, locked recipes, $K=3$).

Fold	HMM (raw)	cGMVAE	HMMGMVAE	cHMM-GMVAE
1	0.003	0.267	0.409	0.579
2	0.001	0.256	0.236	0.396
3	0.003	0.307	0.497	0.675
4	0.208	0.299	0.404	0.649
Mean	0.054	0.282	0.387	0.575

In Table 1, cHMM-GMVAE exceeds cGMVAE on every fold for the temporal prior, with the largest margins on folds 3 and 4 (+0.37 and +0.35 NMI) and the smallest on fold 2 (+0.14), where Bern montages mix with held-out mice from all sites. At matched $T=64$, cHMM-GMVAE also exceeds HMMGMVAE on every fold (mean prior NMI 0.575 versus 0.387). Overall, unsupervised models recover coarse Wake/NREM/REM structure on held-out mice across laboratories, well above chance. REM is the main error mode against

expert macro references used for evaluation, which suggests that three macro classes remain an incomplete description of the signal. On fold 3, Hungarian-aligned macro predictions match expert Wake and NREM well, with REM as the main error mode (Fig 2A). Wake↔NREM switching and REM entry and exit are visible in the transition matrix (Fig 2B).

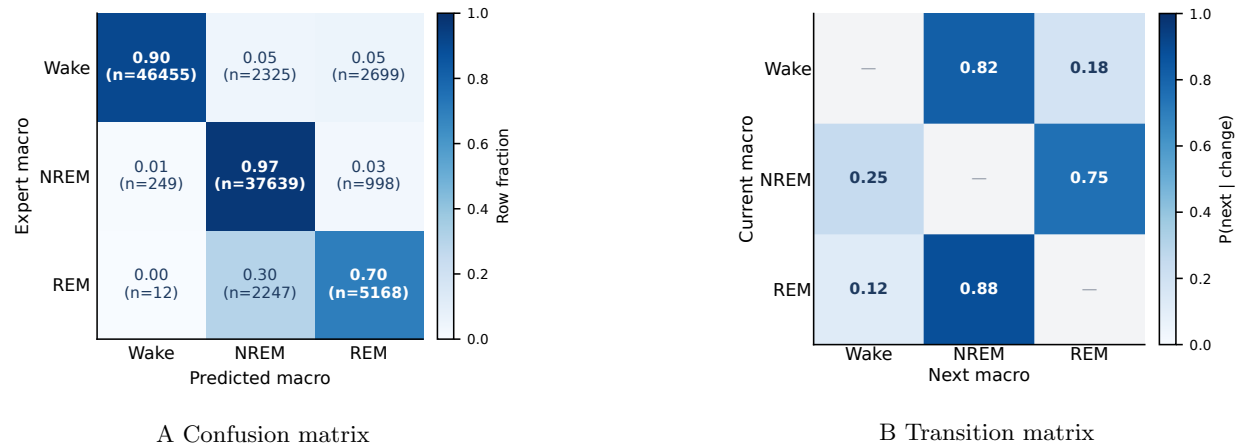


Fig 2. Macro alignment on joint holdout fold 3 (cHMM-GMVAE, best seed, $K=3$). **A**, Row-normalised confusion matrix after Hungarian alignment (rows: expert macro, columns: predicted macro). **B**, Change-only transition matrix $P(\text{next} | \text{current})$ on predicted macros (diagonal omitted).

4.2 LABORATORY-SPECIFIC TRANSFER

Table 2 compares joint population training with within-laboratory training on the *same* held-out mice from each site under leave-mice-out holdout at $K=3$. For joint training, NMI and macro-aligned accuracy are computed on laboratory-specific holdout subsets only, not pooled across sites.

Table 2. Fair per-laboratory holdout comparison (mean over four folds, $K=3$). Joint training uses all laboratories but reports metrics on held-out mice from the listed site only. Within training trains and evaluates inside the same site. Deep models share locked per-laboratory preprocessing and matched hyperparameters (Appendix B.1). They differ in embedding use, sequence length ($T=1$ vs. $T=64$), and prior type. HMM (raw) uses a time-domain pipeline without the shared spectral CNN front-end. Best-of-three prior NMI and macro-aligned accuracy are averaged across folds.

Lab	Model	Joint NMI	Joint Acc.	Within NMI	Within Acc.
Lab 2	HMM (raw)	0.179	0.665	0.004	0.500
	cGMVAE	0.223	0.648	0.549	0.855
	HMMGMVAE	0.039	0.543	0.586	0.865
	cHMM-GMVAE	0.570	0.869	0.596	0.864
Lab 3	HMM (raw)	0.237	0.731	0.029	0.565
	cGMVAE	0.415	0.796	0.516	0.844
	HMMGMVAE	0.586	0.850	0.637	0.897
	cHMM-GMVAE	0.705	0.923	0.738	0.939
Lab 5	HMM (raw)	0.240	0.724	0.001	0.498
	cGMVAE	0.280	0.715	0.476	0.853
	HMMGMVAE	0.483	0.821	0.584	0.882
	cHMM-GMVAE	0.556	0.877	0.567	0.885

Within-lab training exceeds joint training at every site. On held-out Bern mice, for example, cGMVAE prior NMI averages 0.22 under joint training versus 0.55 within-laboratory on the same animals. This

pattern is consistent with residual covariate shift when pooling heterogeneous spectral profiles across sites. Decoder-only subject embeddings absorb amplitude nuisance during reconstruction but do not fully remove equipment and scoring context, so site-specific mismatch can still hurt encoder-based staging at holdout. Macro-aligned accuracy tells a complementary story. On held-out Copenhagen mice, cHMM-GMVAE reaches about 92% under joint training and about 94% within-laboratory (Table 2), and Lyon joint accuracy is about 88%. Those values sit near published expert inter-scorer agreement on rodent EEG/EMG macro epochs (about 95–96% within a laboratory and about 90% across laboratories on non-artifact Wake, NREM, and REM [2]). Against single-rater expert macros used for evaluation, accuracy in the low-to-mid 90% range indicates strong coarse-stage alignment, with REM and transitions as the main remaining disagreement modes.

4.3 SUBSTAGE RESOLUTION AND PHYSIOLOGY

For population substage discovery, expert Wake, NREM, and REM are a post-hoc reference for NMI, not a claim that sleep contains only three dynamical regimes. Rodent sleep biology motivates finer partitions than three macros: NREM may split into light and deep substages (lower δ with sparse spindles versus dominant synchronized slow waves, analogous to human N1–N3), Wake into quiet and active substates, REM occasionally into phasic and tonic components, plus short-bout transition buckets and measurement-artefact epochs. We select mixture count K from mean normalised prior log-density on the population sweep, measuring how well the encoder and temporal mixture prior fit validation latents under the unsupervised checkpoint rule. Macro prior NMI is computed only after that choice and is not used to pick K .

Fig 3 sweeps K on holdout fold 4 for generalisation and on the full cohort for substage discovery. On holdout fold 4, mean macro NMI peaks at $K=3$ and falls at finer resolution, while normalised prior log-density does not track macro NMI (Fig 3A). On the full cohort, normalised prior log-density reaches its maximum at $K=8$ among completed runs (Fig 3B). Mean prior NMI falls as K increases beyond the expert three-class grid, as expected when coarse Wake, NREM, and REM labels cannot reward within-macro subdivisions. We adopt $K=8$ from population prior fit and physiology, not from holdout macro NMI. Population sweeps at $K=3, 11, 13$, and 14 did not finish with usable validation metrics and are omitted from panel B. Other K values use one to five seeds per mixture count, with one collapsed seed excluded from the likelihood aggregate at $K=6$. Holdout physiology at $K=6–12$ is informative but under-powered: only six held-out mice from three laboratories, with high cross-animal variability, so substage occupancy and spectra do not stabilise reliably fold to fold. We therefore characterise substage physiology on all twenty mice under the locked population sweep, retaining Artifact epochs unlike the fold 4 holdout sweep. Expert labels enter only for post-hoc NMI and physiology review. Latent PCA and physiology grids at $K=8$ are clearer with all animals pooled (Figs 4–6). Population panels use the best seed per K under the same unsupervised checkpoint rule ($K=8$ currently has one completed seed).

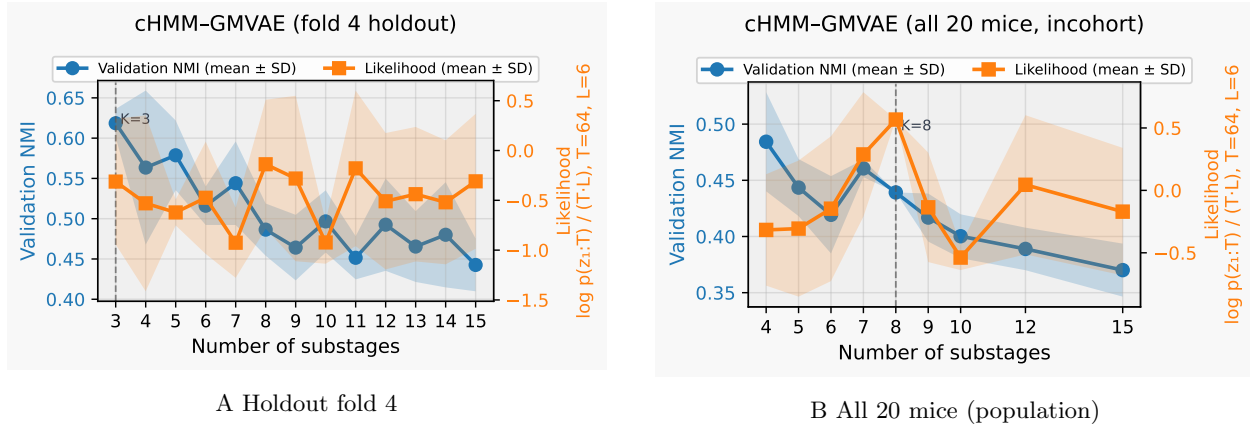


Fig 3. Substage resolution sweeps. Mean validation prior NMI (blue) and mean normalised prior log-density (orange) versus mixture count K . Shaded bands show mean $\pm$ SD across seeds. Collapsed seeds are excluded from the likelihood aggregate only. A, Joint leave-mice-out holdout on fold 4 (six mice, Artifact epochs removed, $K=3-15$). B, Population sweep on the full cohort (Artifact epochs retained, completed $K \in \{4, 5, 6, 7, 8, 9, 10, 12, 15\}$). Prior log-density peaks at $K=8$ on the population sweep. Mean prior NMI falls as K exceeds the three-class expert grid on both panels.

This dual-axis pattern matches likelihood-based model selection against a coarse expert reference. Expert labels supply only three categories while the model assigns K mixture indices, so subdividing one macro into several substages can be physiologically meaningful yet lower NMI because the reference cannot score within-macro structure. Higher K also raises entropy in the predicted label histogram unless every new state sharpens macro alignment. Falling NMI alongside rising prior log-density is therefore expected rather than evidence that high- K mixtures failed.

Figs 4–6 summarise the proposed population map at $K=8$, chosen where normalised prior log-density peaks on the cohort (Fig 3B), with eight active substages and separable physiology. All eight components are occupied despite prior NMI 0.44, illustrating splits within Wake and NREM rather than macro misassignment.

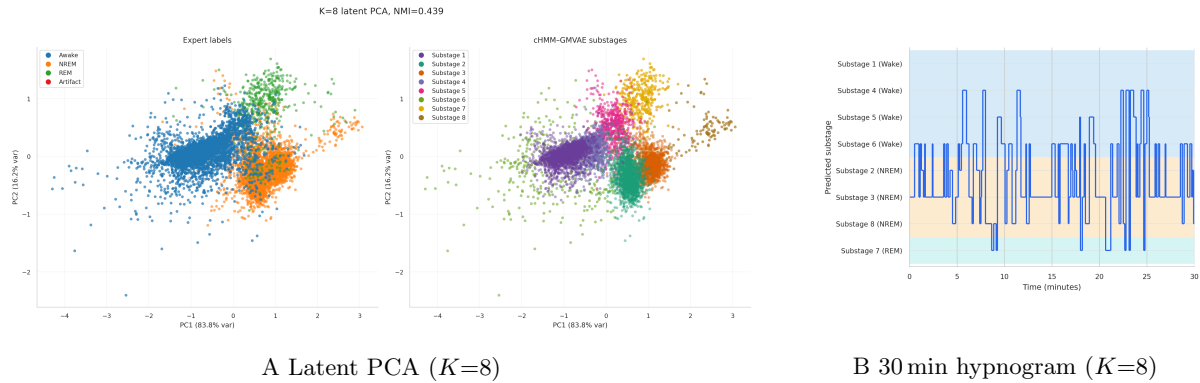


Fig 4. Latent separation and temporal structure at $K=8$ (population sweep, best seed). A, Principal component analysis (PCA) of encoder means: expert Wake/NREM/REM versus predicted substages. B, Example 30 min hypnogram with macro-group background bands.

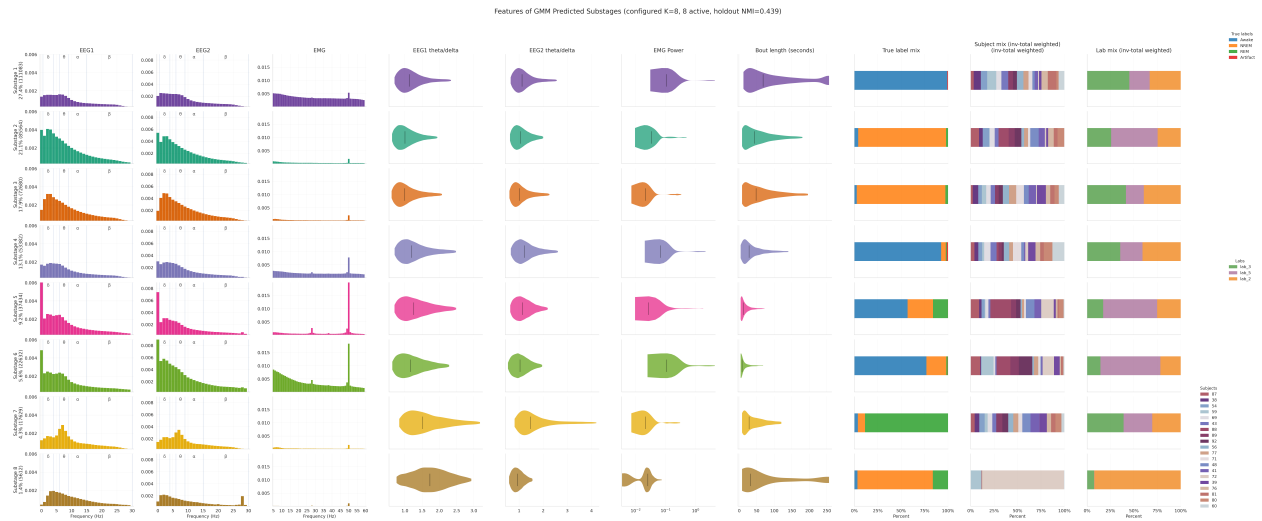


Fig 5. Population-level substage physiology at $K=8$ (locked cHMM-GMVAE, all twenty mice, best seed). Per-substage EEG/EMG power spectra, θ/δ , bout length, expert macro composition, and subject mix. Wake splits into three substages (S1, S4, S6), NREM into two cores (S2, S3), REM consolidates in S7, S5 is a short-bout REM-NREM transition, and S8 is subject-specific.

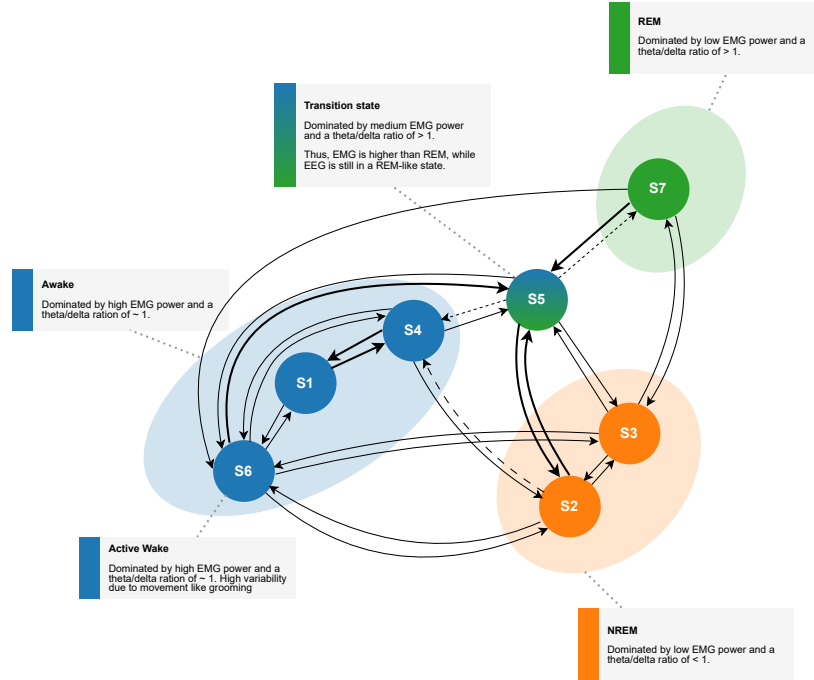


Fig 6. Proposed organisation of interpretable $K=8$ substages into stable Wake, NREM, and REM clusters and short-lived transition buckets (population sweep, best seed; **S8** omitted as subject-specific). Node layout approximates relative substage placement in the cHMM-GMVAE latent space (Fig 4A). Directed edges follow the change-only transition matrix (Appendix D.1); only transitions with $p > 0.05$ are shown. Primary ($p > 0.3$), secondary ($0.1 < p \leq 0.3$), and tertiary ($0.05 < p \leq 0.1$) edges use large black, smaller black, and small grey arrows, respectively. Solid node colours mark stable Wake, NREM, and REM substates; gradients mark transition buckets.

At $K=8$ (Figs 4–6), Wake splits into three substates and NREM into two. **S1** is expert-Wake with elevated EMG and the longest median bouts in the map. **S2** and **S3** are the two NREM substates with separable depth in spectra and balanced occupancy across laboratories and subjects. **S4** is a second Wake bucket that carries the highest fraction of expert Artifact epochs among the eight components. **S5** sits between REM and Wake with lower EMG than the other Wake-dominated rows and short bout lengths, consistent with a REM–Wake transition. **S6** is active Wake with very high EMG and very short bouts, plausibly capturing movement epochs such as mouse grooming. **S7** is the consolidated REM cluster. **S8** is excluded from interpretation because occupancy is concentrated on one mouse at a single site. The remaining seven substates are distributed across all three laboratories and most animals, and are therefore read as the reliable population map. After review of spectra, bout lengths, and transition structure with sleep researchers at the Kornum laboratory, the retained map reads as three Wake substates, two NREM cores, one REM core, one REM–Wake transition stage, and one subject-specific outlier rather than eight equally named sleep stages. Transition structure at $K=8$ is in Appendix D.1. Across EMG rows in Fig 5, persistent peaks near 50 Hz and 28 Hz match electromagnetic interference on unshielded electrodes (50 Hz mains in Denmark and Europe, with additional line-frequency pickup). The schematic summary in Fig 6 groups the retained substates by dominant macro physiology.

At $K=12$ (Appendix D.7), prior log-density does not exceed the $K=8$ peak (Fig 3B) and latent PCA fragments further without a cleaner macro partition. REM still collapses to one dominant substage rather than separating into phasic and tonic components, so we treat twelve substates as an exploratory upper bound rather than a primary map.

Additional population PCA and t-SNE panels at other mixture counts are in Appendix D (including

Appendix D.2 at the same K).

4.4 LIMITATIONS

Our agreement metric is prior NMI against expert macro labels. Those labels are single-rater and disagree across MSSV sites, especially for REM and transitions [1]. Within the same experimental design, cross-laboratory holdout remains harder than within-laboratory training on the same mice (Table 2). For substages, the holdout sweep uses fold 4 only with Artifact epochs removed, whereas population K sweeps train on all twenty mice with Artifact epochs retained and support physiology interpretation rather than leave-mice-out generalisation claims. Headline NMI further uses the best of three seeds per fold, which is optimistic relative to seed means. Proposed substages at $K=8$ are interpreted from unsupervised physiology and transitions and should be read as hypothesis-generating rather than as validated clinical stage names.

5 CONCLUSION

Cross-laboratory mouse sleep can be staged without labels or per-mouse calibration, but only coarsely today. Temporal mixture priors improve Wake/NREM alignment on held-out mice relative to a static mixture prior, with mean prior NMI 0.575 versus 0.282 (+0.29 over folds). Expert labels evaluate coarse alignment only. On the full cohort, an eight-substage map at $K=8$ combines the strongest population prior fit with separable physiology and transition structure, suggesting structure beyond Wake, NREM, and REM alone. Together, these results support label-free cross-laboratory staging and a proposed substage map beyond three expert macro classes, without requiring harmonised scoring across sites.

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A DATA AND COHORT

A.1 MSSV COHORT AND QUALITY SCREENING

MSSV (OpenNeuro ds006366) [7] comprises invasive EEG and neck EMG from 92 mice across five European laboratories, with expert Wake, NREM, REM, and Artifact labels on a 4 s grid for evaluation only. This report analyses laboratories 2 (Bern), 3 (Copenhagen), and 5 (Lyon): healthy wild-type sites with different electrode layouts, excluding laboratories 1 and 4 (pathology-specific or single-electrode cohorts).

Quality screening retains mice with low EEG drift and EMG that separates Wake from NREM. Drift is quantified as relative 0.5–2 Hz power ≤ 0.2 of 0.5–30 Hz on each retained electrode. Wake/NREM separability uses a root-mean-square ratio ≥ 2 on neck EMG. The screened cohort comprises 20 mice (6 Bern, 10 Copenhagen, 4 Lyon) with 42 recording runs and ~ 453 h of EEG/EMG ($\sim 4.1 \times 10^5$ artefact-free 4 s epochs after screening). Training mice never appear in the validation fold for a given split (Appendix A.4). No new animal procedures were performed.

Quality-screened cohort inventory. Per-mouse recording runs, hours, and 4 s epoch counts (MSSV annotation grid).

Mouse	Site	Runs	Hours	Epochs (4 s)
sub-071	Bern (2)	2	12	10,800
sub-072	Bern (2)	2	12	10,800
sub-076	Bern (2)	2	12	10,800
sub-077	Bern (2)	2	13	11,700
sub-080	Bern (2)	2	12	10,800
sub-081	Bern (2)	2	12	10,800
sub-038	Copenhagen (3)	2	48	43,201
sub-039	Copenhagen (3)	3	12	10,836
sub-041	Copenhagen (3)	3	12	10,803
sub-043	Copenhagen (3)	1	24	21,600
sub-048	Copenhagen (3)	3	12	10,804
sub-054	Copenhagen (3)	2	48	43,194
sub-056	Copenhagen (3)	1	23.6	21,196
sub-059	Copenhagen (3)	1	24	21,600
sub-060	Copenhagen (3)	1	8	7,212
sub-069	Copenhagen (3)	1	24	21,600
sub-087	Lyon (5)	6	72	64,788
sub-088	Lyon (5)	2	24	21,595
sub-089	Lyon (5)	2	24	21,596
sub-092	Lyon (5)	2	24	21,595
Total (20 mice)		42	452.6	407,320

A.2 PREPROCESSING AND SPECTRAL INPUTS

Montages are harmonised to a parietal-frontal EEG differential plus neck EMG. Bern (laboratory 2) provides four EEG contacts (EEG1/2 parietal, EEG3/4 frontal); we use EEG1 and EEG4 to match the single parietal-frontal pairing at laboratories 3 and 5 (EEG1 parietal, EEG2 frontal). Recordings are segmented into 4 s epochs at 128 Hz. Artifact epochs are removed. Each epoch is a log-power spectral map $\mathbf{x}_t \in \mathbb{R}^{C \times F}$, batched into contiguous sequences for temporal models without hand-crafted expert features.

Per-laboratory channel sets were fixed from incohort ablation winners (full sweep outcomes in Appendix C.3). Spectral passbands are summarised in Appendix A.3.

A.3 SPECTRAL BANDS AND FREQUENCY CONTENT

Each 4 s epoch comprises 512 samples at 128 Hz sampling rate. The CNN input is a log-power spectrum from an unwrapped real FFT (frequency resolution 0.25 Hz; Nyquist 64 Hz). Band limiting is applied in the frequency domain on the complex spectrum before log-power: a null lower cutoff retains all bins from DC

upward, and an upper cutoff zeroes bins above the configured limit. The model therefore receives the full spectral shape inside each passband, not a small vector of hand-defined sleep bands.

Locked per-laboratory passbands (holdout protocol).

Site	EEG channels	EEG passband	EMG passband
Bern (lab 2)	EEG1, EEG4	0–30 Hz	3–100 Hz (≤ 64 Hz Nyquist)
Copenhagen (lab 3)	EEG1, EEG2	0–20 Hz	5–60 Hz
Lyon (lab 5)	EEG1, EEG2	0–20 Hz	5–60 Hz

Diagnostic and physiology bands. Quality screening compares relative 0.5–2 Hz EEG power to 0.5–30 Hz total (Appendix A.1). For substage physiology panels (Fig 4 and Fig 5) and summary tables, log-power spectra are aggregated into interpretable bands on the same frequency grid: delta 0.5–4 Hz, theta 4–8 Hz, and broadband EMG power summed over the retained EMG passband. Reported θ/δ ratios use these bands.

Bern uses no post-normalisation on log-power spectra. Copenhagen and Lyon apply run-wise z-scoring after band limiting: for each recording run, each channel’s log-power spectrum is scaled by the mean and standard deviation computed across all 4 s epochs in that run (post-normalisation). These limits were chosen from incohort ablations (Appendix C.3) and held fixed for all cross-laboratory holdout runs.

A.4 CROSS-VALIDATION PROTOCOL

Folds were designed so each site contributes holdout mice every fold while balancing validation load: fold 1 is epoch-heaviest ($\sim 32\%$ of cohort epochs in validation, driven by long Copenhagen and Lyon recordings) whereas folds 2–4 hold out 21–25% of epochs each. We use four folds because Lyon contributes only four quality-screened mice, which is the maximum fold count for strict mouse-level holdout. sub-069 is assigned to fold 2 rather than fold 4 so Copenhagen does not contribute four simultaneous holdout mice on the substage-analysis fold. Each VAE configuration trains three random seeds per fold. Reported metrics use the best prior NMI among seeds.

Four-fold leave-mice-out assignments. Top: held-out mice per site and fold. Bottom: validation versus training pool sizes (joint scope trains on all non-held-out mice across sites).

Fold	Site	Held-out mice	Runs	Hours	Epochs
1	Bern (2)	sub-071	2	12	10,800
	Copenhagen (3)	sub-038, sub-039	5	60	54,037
	Lyon (5)	sub-087	6	72	64,788
2	Bern (2)	sub-072, sub-076	4	24	21,600
	Copenhagen (3)	sub-041, sub-048, sub-069	7	48	43,207
	Lyon (5)	sub-088	2	24	21,595
3	Bern (2)	sub-077	2	13	11,700
	Copenhagen (3)	sub-043, sub-054	3	72	64,794
	Lyon (5)	sub-089	2	24	21,596
4	Bern (2)	sub-080, sub-081	4	24	21,600
	Copenhagen (3)	sub-056, sub-059, sub-060	3	55.6	50,008
	Lyon (5)	sub-092	2	24	21,595

Fold	Val mice	Val runs	Val hours	Val epochs	Train mice	Train epochs	Val (%)
1	4	13	144	129,625	16	277,695	31.8
2	6	13	96	86,402	14	320,918	21.2
3	4	7	109	98,090	16	309,230	24.1
4	6	9	103.5	93,203	14	314,117	22.9

B TRAINING, CHECKPOINTS, AND METRICS

B.1 LOCKED HYPERPARAMETERS

Latent dimension $d=6$ (matched to thesis-scale mixture models), embedding dimension 8, learning rate 1.3×10^{-3} , batch size 128, 200 epochs, three random seeds per fold. Sequence length $T=64$ epochs (~ 4.3 min); sticky self-transition weight $\kappa=0.92$. CNN channel widths [32, 64, 128, 128], kernel sizes [7, 5, 5, 3], strides 2; MLP widths [512, 256, 128] (encoder) and [128, 256, 512] (decoder).

B.2 INCOHORT ABLATION RATIONALE

Per-laboratory spectral bandpass, montage, post-normalisation, and MLP width were fixed from incohort sweeps before any holdout run (Appendix A.2, Appendix C.3). These runs pool all mice within a site and estimate a performance *ceiling*; they do not use leave-mice-out folds.

B.3 SENSITIVITY CHECKS

On joint holdout fold 4, sequence length $T=64$ with warm sticky training reached prior NMI ~ 0.65 (best seed), compared with ~ 0.17 for $T=32$ without the warm schedule. On laboratory 3 incohort, warm-started HMM-GMM reached best-of-three prior NMI 0.734 versus 0.694 for a cold HMM-GMM start. Locked settings prioritise reproducibility across holdout runs.

B.4 WARM PRIOR SCHEDULE (CHMM-GMVAE)

Epochs 1–17: standard Gaussian KL on the encoder. Epoch 18: K -means on encoder means initialises GMM emissions. Epochs 18–36: independent mixture KL. From epoch 37: sticky HMM-GMM transitions ramp in over 18 epochs. β is 0 for the first ten epochs, then ramps linearly from 0.01 to 1.0.

B.5 STAGING ON ENCODER MEANS

At holdout, emissions use μ_t rather than sampled $\mathbf{z}_t$. The posterior mean can bias emission scores slightly relative to stochastic latents but reduces Monte Carlo variance in Viterbi decoding and improves reproducibility across seeds once the encoder is trained.

B.6 UNSUPERVISED CHECKPOINT SCORE

Expert labels are not used to select weights during training. Let $\tilde{\omega}(\varepsilon)$ denote validation log-likelihood (sequence-normalised negative ELBO) and $H_{\text{norm}}(\hat{Y})$ the entropy of the predicted state histogram normalised by $\log K$ (0 = collapse to one state, 1 = uniform occupancy). Following the thesis composite criterion [10],

$$S(\varepsilon) = \beta_{\text{ckpt}} \tilde{\omega}(\varepsilon) + (1 - \beta_{\text{ckpt}}) H_{\text{norm}}(\hat{Y}), \quad \beta_{\text{ckpt}} = 0.9. \quad (8)$$

The epoch maximising $S(\varepsilon)$ after a 5% warmup is retained per seed. Prior NMI against expert macros is computed only *after* checkpoint selection for reporting.

B.7 MACRO-ALIGNED ACCURACY AND LABEL SWITCHING

Macro accuracy maps each predicted cluster to its dominant expert macro label (Hungarian alignment when $K=3$), then reports the fraction of epochs matching Wake/NREM/REM. Accuracy can exceed chance and diverge from NMI when $K > 3$ substates collapse onto the same macro label, when too few clusters are active, or when clusters are macro-pure but permuted. Mixture components are identifiable only up to permutation; NMI is invariant to permutations, but cluster indices differ across seeds unless post-hoc alignment is applied for tables and figures.

B.8 RAW TIME-DOMAIN HMM BASELINE

Classical HMMs on raw MSSV EEG/EMG voltages provide a time-domain reference without a shared spectral CNN front-end. On the locked cross-lab protocol, joint HMM (raw) prior NMI averaged 0.054 across four folds (Table 1), motivating the spectral deep-latent models used throughout this report.

B.9 SEED AGGREGATION

Each fold trains three seeds; holdout prior NMI is evaluated at each seed’s $S(\varepsilon)$ checkpoint; the maximum across seeds is the headline fold score (Tables 1–2). All four folds favour cHMM–GMVAE over cGMVAE under the locked recipe.

B.10 PRIOR LOG-LIKELIHOOD IN K-SWEEPS

K-sweep panels (Fig 3) report mean normalised prior log-density alongside prior NMI. For cGMVAE ($T=1$), density is per 4s epoch; for HMMGMVAE and cHMM–GMVAE ($T=64$), density is the HMM forward marginal per sequence window, normalised by $T \cdot L$ for comparability on the dual-axis plots.

C PRIOR MODELS AND PROTOCOL ABLATIONS

C.1 HMM–GMM PRIOR (EQUATIONS)

Each discrete latent state $s_t \in \{1, \dots, K\}$ emits a diagonal Gaussian in $\mathbb{R}^d$. The per-step emission log-density is

$$\log b_k(\mathbf{z}_t) = \log \mathcal{N}(\mathbf{z}_t \mid \mathbf{m}_k, \text{diag}(\mathbf{v}_k)). \quad (9)$$

Sticky transitions use self-transition weight $\kappa=0.92$. The marginal log-likelihood of a latent window is

$$\log p(\mathbf{z}_{1:T}) = \log \sum_{s_{1:T}} \pi_{s_1} \prod_{t=2}^T A_{s_{t-1}, s_t} \prod_{t=1}^T b_{s_t}(\mathbf{z}_t), \quad (10)$$

which enters the training KL term $\mathcal{L}_{\text{KL}}$ in Eq. (5). Warm-start phases and CNN widths are in Appendix B.4 and Appendix B.1. Controlled synthetic datasets in the MSc thesis validated MAR-HMM optimisation before MSSV experiments (thesis Ch. 3). That validation is not repeated here.

C.2 DECODER-ONLY VS ENCODER+DECODER CONDITIONING

The MSc thesis default fed subject embeddings to *both* encoder and decoder on laboratory 3 only [10]. Before holdout, we compared decoder-only and encoder+decoder paths on locked incohort recipes at all three sites ($K=3$, three seeds, Appendix C.2):

Site	Decoder-only	Encoder+decoder	Best-of-three prior NMI
Bern (lab 2)	0.593	0.578	Decoder-only
Copenhagen (lab 3)	0.737	0.621	Decoder-only
Lyon (lab 5)	0.534	0.430	Decoder-only

All three sites favoured decoder-only conditioning on incohort prior NMI. The thesis headline (~ 0.70 population NMI) used encoder+decoder on laboratory 3 with expert features and leave-one-subject-out evaluation [10], which is not directly comparable to these three-lab spectral incohort runs. Holdout fold-level encoder+decoder comparisons were not run under the locked paper protocol. The main line uses decoder-only throughout.

C.3 INCOHORT PERFORMANCE CEILING

Before cross-laboratory holdout, we ran extensive within-laboratory sweeps over preprocessing, CNN–VAE architecture, prior tier (static GMM vs. HMM–GMM vs. warm sticky HMM–GMM), and sequence length T (Table C.3). All mice in a laboratory were pooled for training and validation tuning (no leave-mice-out). Reported values are best-of-three prior NMI at $K=3$ with decoder-only conditioning. These runs fixed the per-site recipes in Appendix A.2 and estimate an upper performance bound when site-specific tuning is allowed. They are *not* the locked holdout protocol in Table 2. Macro accuracy was not logged systematically in the incohort sweeps.

Bern (laboratory 2): cGMVAE reached 0.593 with EEG1+EEG4, no post-normalisation, and EEG 0–30 Hz. Temporal priors did not beat the static mixture at $T=1$ (best cHMM–GMVAE 0.571 with warm prior and 50 Hz notch). Copenhagen (laboratory 3): cGMVAE peaked at 0.737; warm cHMM–GMVAE at $T=64$

matched within 0.003 (0.734). Lyon (laboratory 5): cHMM-GMVAE with EMG-wide notch preprocessing at $T=32$ reached 0.640 versus 0.534 for cGMVAE.

Incohort ceiling prior NMI. Best-of-three prior NMI and winning configuration per laboratory and model.

Lab	Model	NMI	Winning lock (summary)
2	cGMVAE	0.593	EEG1+EEG4; no post-norm.; EEG 0–30 Hz; EMG 3–100 Hz
2	cHMM-GMVAE	0.571	Warm prior; 50 Hz notch
3	cGMVAE	0.737	Post-norm.; EEG 0–20 Hz; MLP 512–256–128
3	cHMM-GMVAE	0.734	Warm HMM-GMM; $T=64$; same preprocessing
5	cGMVAE	0.534	Post-norm.; EEG 0–20 Hz; MLP 512–256–128
5	cHMM-GMVAE	0.640	EMG wide + notch; $T=32$

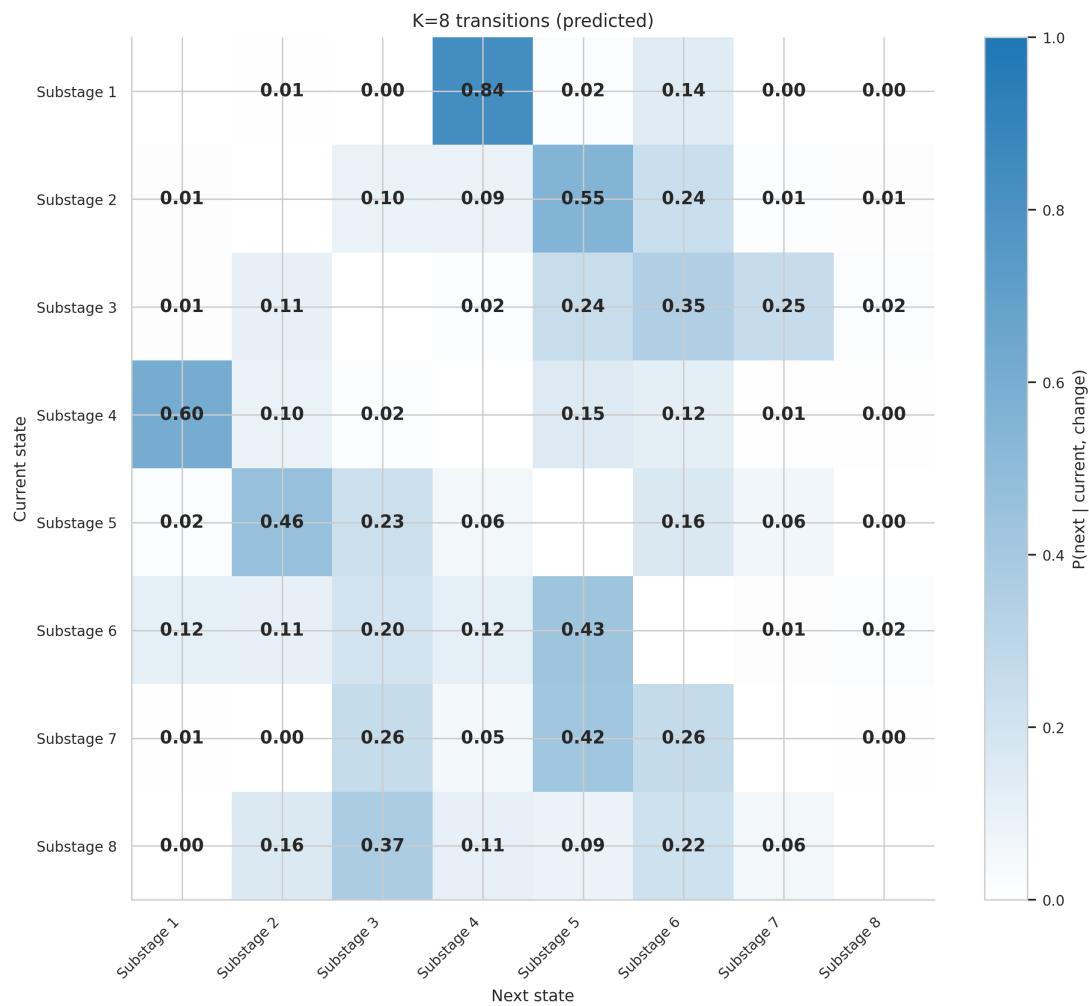
Ceiling reference only. Holdout within-lab NMI and accuracy under the same locked recipes appear in Table 2. Joint holdout remains below these ceilings because leave-mice-out validation is strictly harder than pooling all laboratory mice during tuning. The ceilings also show that cGMVAE can reach much higher prior NMI under site-tuned incohort settings (for example 0.737 at Copenhagen) than under the unified holdout recipe (joint mean 0.282). That gap is expected because holdout locks one preprocessing and training recipe across cGMVAE, HMMGMVAE, and cHMM-GMVAE for fair prior comparison rather than maximising the static mixture alone. A useful follow-up is to select cGMVAE-specific hyperparameters from incohort sweeps at $T=1$ and re-run joint holdout, to see how much of the incohort ceiling transfers under leave-mice-out and whether cHMM-GMVAE retains its margin.

D SUPPLEMENTARY FIGURES

Population substage dynamics at the operating mixture count ($K=8$) are in Appendix D.1; latent embeddings at the same K are in Fig 4 and Appendix D.2.

D.1 POPULATION SUBSTAGE TRANSITIONS AT $K=8$

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Population $K=8$ change-only transition matrix (locked cHMM-GMVAE, all twenty mice, best seed). Rows are current substates, columns are next substates; diagonal omitted ($P(\text{next} \mid \text{current})$). **S7** (REM) exchanges brief epochs with **S5** (REM–Wake transition). NREM cores **S2** and **S3** cycle within NREM. Wake substates **S1**, **S4**, and **S6** form separate within-Wake loops (**S8** omitted as subject-specific). Cited in Results alongside Figs 4–6.

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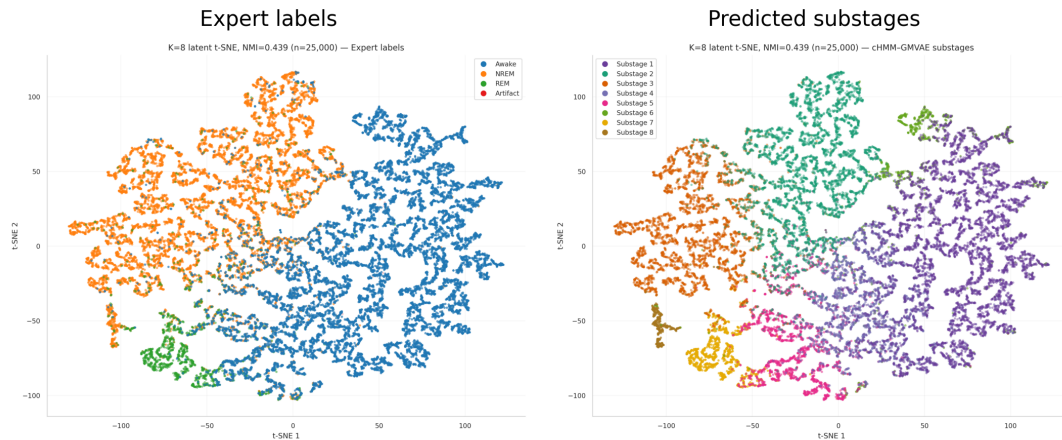
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D.2 POPULATION T-SNE AT $K=8$

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Population $K=8$ latent t-SNE (best seed, shared embedding on a fixed subsample of encoder means). **Left**, expert Wake/NREM/REM. **Right**, predicted substages. Complements Fig 4A (PCA at the same K).

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D.3 POPULATION SUBSTAGE PCA ($K=4, 7, 9, 12$)

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Locked cHMM-GMVAE, all 20 mice (incohort), best seed per K (main text: $K=8$ latent panels in Fig 4, physiology and biological summary in Figs 5 and 6, transitions in Appendix D.1). Each panel below shows PCA of encoder means with expert Wake/NREM/REM versus predicted substages at other mixture counts cited in the main Results.

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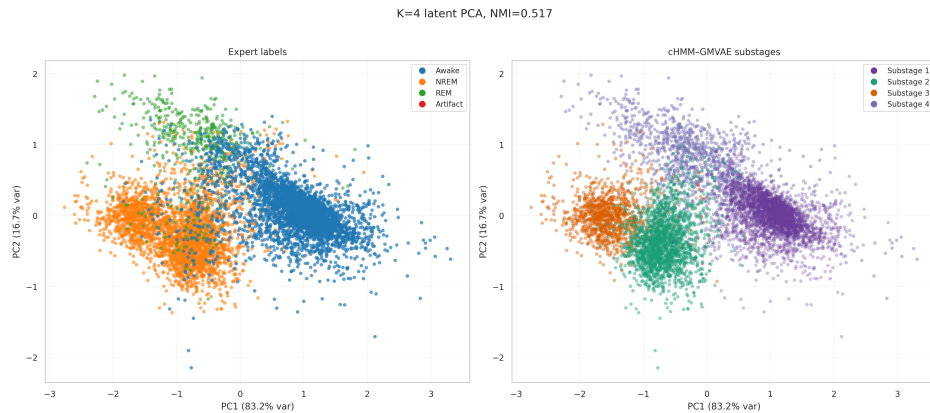
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D.4 POPULATION PCA AT $K=4$

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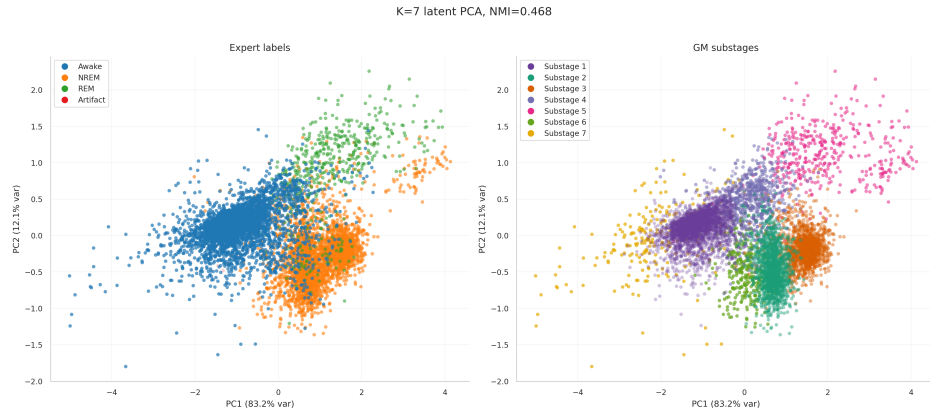
Population $K=4$ (best seed, prior NMI 0.517). Latent PCA of expert Wake/NREM/REM versus predicted substages, a coarser map before $K=8$.

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D.5 POPULATION PCA AT $K=7$

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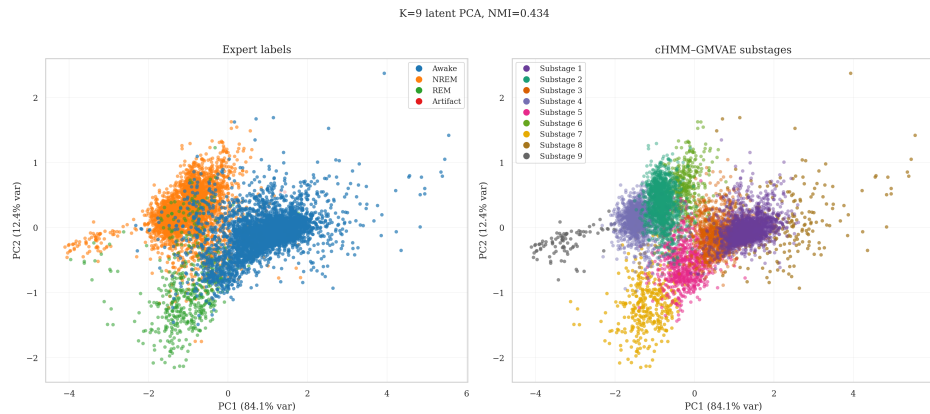
Population $K=7$ (best seed, prior NMI 0.468). Latent PCA of expert macros versus predicted substages at intermediate resolution between $K=4$ and the main $K=8$ map.

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D.6 POPULATION PCA AT $K=9$

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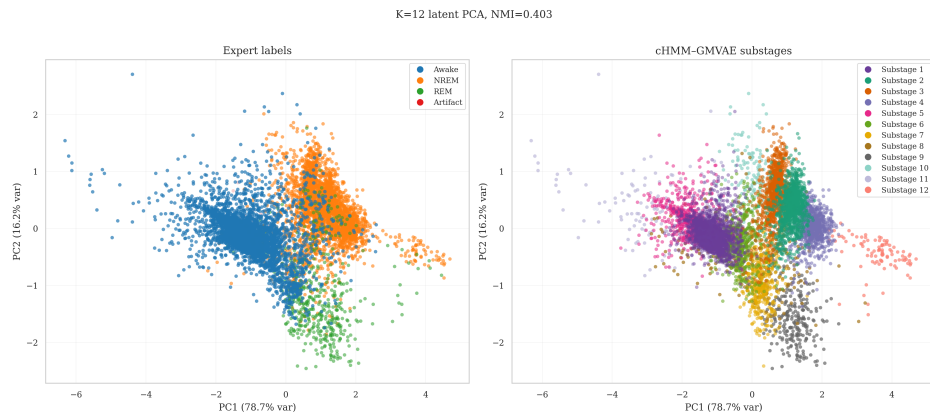
Population $K=9$ (best seed). Latent PCA at higher resolution than the main $K=8$ map. Wake subdivides further without a corresponding gain in prior log-density.

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D.7 POPULATION PCA AT $K=12$

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Population $K=12$ (best seed). Exploratory upper-bound resolution. Several substates are unstable or subject-specific compared with $K=8$.

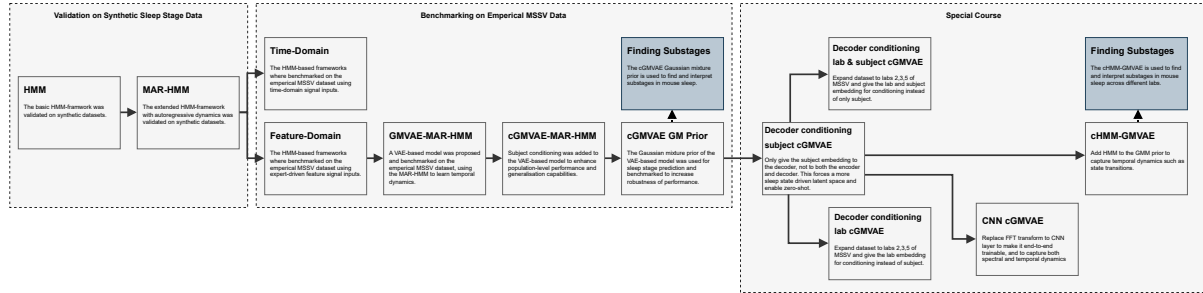
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E RESEARCH STORY MAP

Roadmap from synthetic and incohort validation through cross-laboratory holdout to population substage discovery at $K=8$ (Fig E.1).

E.1 STORY MAP



Research story map: ordered analyses, main-text figures, and appendix panels cited in this report. Holdout macro staging (Figs 3A, 2; Tables 1–2) precedes population K -sweeps (Fig 3B) and the $K=8$ substage map (Figs 4–6; Appendix D.1).